

# Effect of inoculation with native and commercial arbuscular mycorrhizal fungi on growth and mycorrhizal colonization of olive (*Olea europaea* L.)

Haroun Chenchouni<sup>a,b,\*</sup>, Mohamed Nacer Mekahlia<sup>a</sup>, Arifa Beddiar<sup>c</sup>

<sup>a</sup> Department of Natural and Life Sciences, Faculty of Exact Sciences and Natural and Life Sciences, University of Tebessa, 12002 Tebessa, Algeria

<sup>b</sup> Laboratory of Natural Resources and Management of Sensitive Environments 'RNAMS', University of Oum-El-Bouaghi, 04000 Oum-El-Bouaghi, Algeria

<sup>c</sup> Department of Biology, Faculty of Natural and Life Sciences, University of Badji Mokhtar, 23000 Annaba, Algeria

## ARTICLE INFO

### Keywords:

Arbuscular mycorrhizae  
Biotic plant responses  
Controlled inoculation  
Mycorrhizal parameters  
Olive tree *Olea europaea*  
Plant-AMF interactions  
Native soil fungi  
*Glomus* spp.

## ABSTRACT

In order to improve the production of olive (*Olea europaea* L.), plantlets of Variety Ferkani were inoculated with three arbuscular mycorrhizal fungal (AMF) spores (*Glomus* sp.1, *Glomus* sp.2, *Septoglomus constrictum*) isolated from a hot-arid zone in Algeria and three commercial AMF (*Rhizophagus intraradices*, *Funneliformis mosseae*, *Gigaspora margarita*). Mycorrhization parameters and nine growth traits were estimated in two-year-old plantlets, then the variation of growth traits following mycorrhizae parameters of various AMF inoculums was tested. The tested AMF species were all infective and effective for the mycorrhization of Ferkani variety. Native species *Glomus* sp.1 gave the best results for mycorrhizal colonization and growth parameters. Indeed, the highest colonization frequency (*F*%) was recorded in plantlets inoculated with *Glomus* sp.1 ( $58 \pm 6.5$  %), whereas the highest values of colonization intensity of in the root system (*M*%) were obtained with *Rhizophagus intraradices* ( $2.45 \pm 1.46$  %) and *Glomus* sp.1 ( $2.26 \pm 0.76$  %). Abundances of arbuscules were low for all AMF inoculums. The controlled inoculation improved the growth of all plantlets compared to the AMF-free plantlets. AMF-inoculated plantlets showed significantly higher growth in terms shoot height, fresh and dry weights of shoot and root, and number of leaves and nodes. Statistical models revealed that all the latter growth parameters including mycorrhizal responsiveness increased significantly when mycorrhizal parameters increased, and that *F*% was most strongly correlated with plant growth compared to other mycorrhizal parameters. Findings of this controlled inoculation research underline the importance of local AMF isolated from the rhizosphere of Ferkani variety in rapidly and extensively infecting the root system and then increasing growth of the host.

## 1. Introduction

Impacts of agricultural practices on the environment are very diverse. The negative impacts of some of these practices affect the integrity of both natural habitats and agro-ecosystems with harmful repercussions on human health (Hillocks, 2012; Hernández et al., 2013). Indeed, intensive use of chemicals in agriculture, including fertilizers and pesticides, may be a solution to low agricultural production. However, in addition to exorbitant costs of this practice, it results in adverse effects on human health and ecosystem functioning (Schreinemachers and Tipraqsa, 2012). The ecological agriculture is a

new way for sustainable agriculture, which aims to use sustainably natural processes and ecosystem functionalities while having a good agricultural yield (Oustani et al., 2015; Boudjabi et al., 2015, 2019). The agri-environment puts forward the question of the debate between sustainable development and chemical agriculture (Kleijn et al., 2004); a debate that has intensified since a great deal of research has focused on the impact of chemical agriculture on the environment (Aktar et al., 2009; Mekahlia et al., 2016; Carvalho, 2017; Bouaroudj et al., 2019).

Olive cultivation is one of the fundamental cultures in the Mediterranean basin. The distributional range of the olive tree (*Olea europaea* L.) has been used to define the Mediterranean climatic area

**Abbreviations:** A%, abundance of arbuscules in the whole root cortex [%]; a%, abundance of arbuscules in the mycorrhized parts of root fragments [%]; AMF, arbuscular mycorrhizal fungi; ANOVA, analysis of variance; F%, frequency of mycorrhizal colonization [%]; GLM, generalized linear model; M%, mycorrhizal colonization of the whole root cortex [%]; m%, mycorrhizal colonization of root fragments [%]; MR, mycorrhizal responsiveness [%]; NL, number of leaves; NN, number of nodes; RFW/SFW, ratio between root fresh weight and shoot fresh weight; RFW, root fresh weight [g]; SDW, shoot dry weight [g]; SFW, shoot fresh weight [g]; SH, shoot height [cm]; SRWC, shoot relative water content [%]

\* Corresponding author at: Department of Natural and Life Sciences, Faculty of Exact Sciences and Natural and Life Sciences, University of Tebessa, 12002 Tebessa, Algeria.

E-mail addresses: [chenchouni@gmail.com](mailto:chenchouni@gmail.com) (H. Chenchouni), [mekahlia79@yahoo.fr](mailto:mekahlia79@yahoo.fr) (M.N. Mekahlia), [fragbed@yahoo.fr](mailto:fragbed@yahoo.fr) (A. Beddiar).

<https://doi.org/10.1016/j.scienta.2019.108969>

Received 31 May 2019; Received in revised form 18 October 2019; Accepted 21 October 2019

0304-4238/ © 2019 Elsevier B.V. All rights reserved.

and delimit the Mediterranean-type regions (Blondel et al., 2010). In terms of area, olive tree farming in Algeria is the most important arboreal fruit sector with almost 40 % of areas (ITAFV, 2013; Chafaa et al., 2019). The number of olive varieties is 48 cultivars, among which the Ferkani variety is endemic to the Aurès region in northeastern Algeria (INRAA, 2006). The severe climatic conditions prevailing in this zone, especially at its southern part near the Sahara Desert (Benabderrahmane and Chenchouni, 2010; Chenchouni, 2010), are affecting negatively the development, stability and yield of olive cultivation (Boughalleb and Hajlaoui, 2011). Low olive yields are also due to poor soil quality, in particular nitrogen and phosphorus (Birhane et al., 2012). In order to cope with these severe environmental conditions and to ensure a constant productivity in the olive tree, the use of biological solutions has often been considered (Porrás-Soriano et al., 2009). Thus, an increasing interest has been attributed to endomycorrhizal fungi, which appear to be the most important telluric organisms to consider and value (Akhtar and Siddiqui, 2008; Meddad-Hamza et al., 2017).

Arbuscular mycorrhizal symbiosis is a mutually beneficial interaction between soil arbuscular mycorrhizal fungi (AMF) and more than 80 % of vascular plant families (Brundrett, 1991; Wang and Qiu, 2006). It is also an important integral part of natural ecosystems (Garg and Chandel, 2010). AMF comprise more than 150 species belonging to the orders Glomerales, Diversisporales, Gigasporales, Archaeosporales and Paraglomerales (Oehl et al., 2011a). Because of their ubiquity and physiological characteristics, AMF play several very important roles in plant ecology, particularly in mineral nutrition (Parniske, 2008; Helgason and Fitter, 2009), salt stress mitigation (Ghazi and Al-Karaki, 2006; Giri et al., 2007), water supply and drought resistance (Nelsen, 1987; Ruiz-Lozano et al., 2012), protection against pathogens (Harrier and Watson, 2004; Akhtar and Siddiqui, 2008), bioremediation (Xavier and Boyechko, 2002; Garg and Chandel, 2010), phyto-hormone production (Johansson et al., 2004), adaptation to various adverse environmental conditions (Miransari et al., 2008; Birhane et al., 2012), and ecosystem and ecological balance (Oehl et al., 2011b; Sivasithamparan et al., 2002).

Even if the potential of these symbiotic fungi is recognized (Xavier and Boyechko, 2002; Harrier and Watson, 2004; Parniske, 2008; Ruiz-Lozano et al., 2012), including for the olive tree (Porrás-Soriano et al., 2009; Mekahlia et al., 2013; Meddad-Hamza et al., 2017), their effective and large-scale applications in agricultural systems remains to be done. Thus, the main objective of this ecophysiological study is to highlight the beneficial effects that represent different AMF inoculations for the olive variety Ferkani. Using olive plantlets obtained by cuttings and having the same genetic characteristics, the study aims at estimating and comparing parameters of endomycorrhizal colonization and also plantlet growth traits for different inoculations involving native (*Septoglomus constrictum*, *Glomus* sp.1 and *Glomus* sp.2) and commercial (*Rhizophagus irregularis*, *Funneliformis mosseae* and *Gigaspora margarita*) AMF. Concurrently, the objective of these root inoculations with different AMF strains is to determine the efficacy and infectivity of these AMF strains for the variety studied. The study investigates the effects mycorrhizal parameters of various AMF species on plant growth traits.

## 2. Materials and methods

### 2.1. Removal of cutting and preparation of plant material

Olive plantlets used in controlled mycorrhization tests were axenically grown cuttings. In order to reduce biological variability, the cuttings used all come from a single, healthy and vigorous adult tree of the Ferkani variety, from a biological tree plantation in the Ferkene region (latitude: 34°29'13.32"N, longitude: 7°28'6.91"E) located southern of the Wilaya 'Provence' of Tebessa (northeastern Algeria). The region of Ferkene is characterized by an arid climate, whereas the

region of Tebessa in general has a semi-arid climate where steppe rangeland vegetation dominates (Neffar et al., 2018; Djellab et al., 2019). The cuttings used were taken during the beginning of spring (March), because this period corresponds to an intense cambial activity which favors the emission of roots. They come from the branches of the base of the tree because they root better and faster than those at the top of the tree. On each cutting, the basal leaves were removed and 4–6 apical leaves were left with 8–12 buds. The cuttings studied measured on average 10–12 cm in length with an average diameter of 0.4 cm. To trigger floral induction, the base of each cutting (2–3 cm) was soaked for 5 s in an auxin (indol butyric acid 'IBA' with 3000–5000 ppm) then in 70 % ethanol, after that left for 10–15 min to dry out.

### 2.2. Cultivation conditions in nebulization greenhouse

The length of stay of the cuttings inside the nebulization greenhouse was 2 months. At the greenhouse setting, cuttings were cultivated on multiplication tables filled with a layer of 12–15 cm thick of perlite, which is an inert substance (pH = 7). This substrate ensures adequate aeration at the base of the cutting and retains just the needed amount of water and well-draining the excess, and thus promotes a good development of the root system. The planting depth was 3–5 cm with a planting density of 400–800 cuttings/m<sup>2</sup>. The other culture conditions included the ambient temperature in the greenhouse that was 20–25 °C during the day and 13–15 °C at night, while the substrate temperature at the base of the cuttings was 18–22 °C. The relative humidity of the air is kept close to saturation (i.e. ~90 %). Misting or irrigation maintained a film of water on the surface of the leaves, i.e. straying for 5–10 s every 10–15 min. Shading inside the greenhouse was 50 % to ensure a high level of rooting. The callus appeared after 18–20 days, while the first roots appeared after the 40th day.

### 2.3. Transplantation of plantlets

At the age of 3 months, and after root emission and shoot development in the nebulization greenhouse, the rooted cuttings were transplanted into 5-liter pots containing sterilized sandy soil from the Ferkene region. Soil was autoclaved twice for a period of 2 h at 120 °C, with 24-h interval between the two sterilizations. Each pot contained one olive plantlets. This approach is applied to enable young plantlets gradually acclimate to the conditions of the external environment. Plantlets were sprayed with distilled water; meanwhile, a nutrient solution was used fortnightly at a rate of 20 mL/pot. The nutrient solution contained the following macronutrients (in meq/pot):  $\text{NH}_4^{4+} = 0.75$ ,  $\text{NO}_3^- = 3.75$ ,  $\text{K}^+ = 1.75$ ,  $\text{Ca}^{2+} = 0.80$ ,  $\text{Mg}^{2+} = 0.30$ ,  $\text{H}_2\text{PO}_4^- = 0.35$ ,  $\text{SO}_4^{2-} = 0.30$  (Tattini, 1990).

### 2.4. Inoculation of plantlets with native and commercial AMF

The inoculation of olive plantlets with native and commercial AMF was carried out when plantlets were transferred into the pots (age ≈ 3 months). The native fungal strains used were collected from the soil of the region of Ferkene, viz. *Glomus* sp.1, *Glomus* sp.2, and *Septoglomus constrictum* (syn. *Glomus constrictum*). While the commercial AMF strains studied were commercially available and have been supplied in the form of granules, spores and roots by Biorize R&D (France), viz. *Rhizophagus intraradices* (syn. *Glomus intraradices*), *Funneliformis mosseae* (syn. *Glomus mosseae*), and *Gigaspora margarita*. The binomial nomenclature of AMF species studied were established following systematic updates of MycoBank 'MB' (Robert et al., 2013; www.mycobank.org). Species synonyms were given to enable comparisons between studies. Uninoculated treatment (control) was also implemented by simply transplanting olive plantlets under the same condition, but with no AMF inoculation. Five repetitions were performed for each treatment (six AMF strains and one control).

## 2.5. Plantlet measurements

After two years of growth, the following parameters were measured on each olive sapling: SH: shoot height [cm]; NL: number of leaves; NN: number of nodes; SFW: shoot fresh weight [g]; SDW: shoot dry weight [g] determined after oven-drying the fresh material at 70 °C for 72 h; SRWC: shoot relative water content [%]; RFW: root fresh weight [g]; RFW/SFW: ratio between root fresh weight and shoot fresh weight; and MR: mycorrhizal responsiveness [%], with  $MR = [SDW \text{ (mycorrhized plantlets)} - SDW \text{ (control plantlets)}] / SDW \text{ (mycorrhized plantlets)}$  (Janos, 2007). In order to obtain the MR of each mycorrhized plantlets, MR was computed based on observed SDW of mycorrhized plantlets, while SDW of control plantlets was averaged.

## 2.6. Determination of arbuscular mycorrhizal colonization

The entire root system of each plantlet was recovered and staining in the laboratory to estimate the level of mycorrhization. The staining technique used is that recommended by Vierheilig et al. (2005), which consisted of recovering the small non-lignified fine roots, cleaning them well in order to eliminate any earth particles, then cutting them into fragments of about 1 cm in length, and putting them first in a water bath at 90 °C in a solution of potassium hydroxide (10 % KOH) for one hour to empty the cells of their cytoplasmic content to improve the observation of the symbiotic fungus. The root fragments were rinsed with water to remove all traces of KOH, they were immersed successively in hydrogen peroxide (H<sub>2</sub>O<sub>2</sub> at 10 vol) for 40 min and in hydrochloric acid (2 %) for 30 min to bleach them. After a new rinse with water was performed, bleached root fragments were stained for one hour using chlorazol black E (0.1 g chlorazol black E + 100 mL distilled water + 100 mL lactic acid + 100 mL glycerol) heated to 90 °C in a water bath (Brundrett et al., 1984). Chemicals were of analytical grade and obtained from Sigma-Aldrich (Germany). The stained roots were kept in labeled pill containers containing glycerol to be observed later.

## 2.7. Estimation of root mycorrhization

The percentage of colonization of the roots was calculated following the method of Trouvelot and Kough (1986), which is a fast-technique reflecting as much as possible the potential and the state of activity of mycorrhizal symbiosis. It consisted in putting fifteen fragments of stained roots between slides and lamellae and observing them under a light microscope. The operation was repeated twice to calculate five mycorrhization parameters using the MycoCalc computer program (<http://www2.dijon.inra.fr/mychintec/>):

- Frequency of mycorrhizal colonization 'F%' which corresponds to the ratio between of endomycorrhizal root fragments and the total number of root fragments examined,
- Mycorrhizal colonization of the whole root system 'M%',
- Mycorrhizal colonization of root fragments 'm%': estimated as the amount of mycorrhized root cortex, relative only to the mycorrhizal root fragments,
- Arbuscule abundance 'A%' that estimated arbuscule richness in the whole root system,
- Abundance of arbuscules in the mycorrhized parts of the root fragments 'a%'.

## 2.8. Statistical analysis

Values of mycorrhizal parameters (F%, M%, m%, A%, a%) were summarized for each native and commercial AMF spore using some descriptive statistics viz. mean, standard deviation 'SD', interquartile range 'IQR', coefficient of variation 'CV', Skewness, Kurtosis). Data related to olive plantlet growth parameters per AMF species were visualized using boxplots. One-way analysis of variance 'ANOVA' was

applied to test the variation of mycorrhizal parameters and plantlet traits between the AMF species tested. Each ANOVA was followed by a Tukey's honest significance test 'HSD' to distinguish homogeneous groups of AMF inoculums. Interrelationships between mycorrhizal parameters and between plantlet traits were tested using Pearson correlations at  $\alpha = 0.05$ . Each correlation matrix, i.e. one for mycorrhizal parameters and another for plantlet traits was displayed using the corrplot package in R (Wei and Simko, 2016). The relationship between each mycorrhizal parameter and the variation of olive growth traits were tested using generalized linear models (GLM). During GLM computations, NL and NN data were fitted to a Poisson distribution and log link, whereas the other plant traits (SH, SFW, SDW, SRWC, RFW, RFW/SFW ratio, and MR) were fitted using Gaussian distribution and identity link function. Then, the effect of each mycorrhizal parameter was determined using type-III likelihood ratio tests for NL and NN and type-III F-tests for the rest of plant traits. Statistical analyses and models were carried out using the software R (R Core Team, 2019).

## 3. Results

### 3.1. Olive mycorrhizal parameters for different AMF strains

The mycorrhizal frequency (F%) in roots of the variety Ferkani ranged from  $32.67 \pm 3.65$  % in *Glomus* sp.2– $58 \pm 6.5$  % in *Glomus* sp.1 (Table 1). Averages of M% varied between  $0.43 \pm 0.11$  % recorded in *Glomus* sp.2 and  $2.45 \pm 1.46$  % in *Rhizophagus intraradices*. The lowest average of m% was noted in *Glomus* sp.2  $1.34 \pm 0.34$  %, whereas its highest mean was recorded in *Glomus* sp.1 with  $3.99 \pm 1.57$  %. For the abundance of arbuscules, A% averaged very low levels that varied between 0 % (total absence) in *Glomus* sp.2 and  $0.05 \pm 0.05$  % in *Glomus* sp.1 and *R. intraradices*. The a% parameter also revealed very low levels, ranging from 0 % in *Glomus* sp.2– $2.24 \pm 2.14$  % in *Glomus* sp.1. The ANOVA showed that the variation in mycorrhization parameters of olive plantlets among the fungal inoculums was significant for the values of F% ( $F_{(5,24)} = 15.70$ ,  $P < 0.001$ ), M% ( $F_{(5,24)} = 4.54$ ,  $P = 0.005$ ) and m% ( $F_{(5,24)} = 2.65$ ,  $P = 0.048$ ). Tukey's HSD test indicated that F% values were significantly higher in both *Glomus* sp.1 and *R. intraradices*. Also, the latter species in addition to *Funneliformis mosseae* showed significantly higher M% compared to other species. The lowest values of F% and M% were marked in *Glomus* sp.2 that was found not different of *S. constrictum*, *F. mosseae* and *G. margarita*. The other AMF species have intermediate positions between the previous species (Table 1). Pearson's correlation tests between mycorrhizal parameters of olive plantlets showed that all correlations were significantly positive (Fig. S1). Out of ten correlation tests, six showed strong correlations ( $r > 0.5$ ), which were observed between F%–M%, F%–m%, M%–m%, M%–A%, m%–A%, and A%–a%.

### 3.2. Variation of Plantlets growth traits between AFM treatments

Results of growth parameters recorded in olive plantlets are expressed in Fig. 1 that shows how much the controlled inoculation improved the growth of the two-year-old plantlets compared to the controls (AMF-free plantlets). One-way ANOVA showed that the AMF species used had a highly significant effect ( $P < 0.001$ ) on the variation of all olive plantlet growth parameters, except for SRWC and RFW/SFW ratio ( $P > 0.05$ ). Regardless of the AMF species tested, Tukey tests revealed that the AMF-inoculated plantlets showed significantly higher MR values compared to the control plantlets. For the growth parameters SH, NL, NN, SFW, SDW, and RFW this increase compared to the control plantlets differed following AMF species. Averages of plant shoot height 'SH' varied between 77.2 cm in *Glomus* sp.2 and 145.6 cm in *Glomus* sp.1, while SH of the control did not exceed on average 54.6 cm. A greater weight gain in the aerial fresh matter was also recorded in the mycorrhizal plants compared to the controls; SFW averages ranged between 422 g in *F. mosseae* and 1123 g in *Glomus* sp.1.

**Table 1**Descriptive statistics and variation of mycorrhizal parameters in olive (*Olea europaea*) saplings inoculated with different AMF species.

| AMF spores inoculated           | Descriptive statistics of mycorrhizal parameters <sup>A</sup>  |      |      |      |          |          | ANOVA <sup>B</sup>  |         | Tukey HSD <sup>C</sup> |
|---------------------------------|----------------------------------------------------------------|------|------|------|----------|----------|---------------------|---------|------------------------|
|                                 | Mean                                                           | SD   | IQR  | CV   | Skewness | Kurtosis | F <sub>(5,24)</sub> | P-value |                        |
|                                 | Mycorrhizal colonization frequency (F%)                        |      |      |      |          |          |                     |         |                        |
| <i>Glomus</i> sp.1              | 58.00                                                          | 6.50 | 6.67 | 0.11 | 0.08     | −0.82    | 15.70               | < 0.001 | "c"                    |
| <i>Glomus</i> sp.2              | 32.67                                                          | 3.65 | 0.00 | 0.11 | −1.29    | 2.91     |                     |         | "a"                    |
| <i>Septoglomus constrictum</i>  | 47.33                                                          | 2.79 | 3.33 | 0.06 | −0.52    | −0.60    |                     |         | "b"                    |
| <i>Rhizophagus intraradices</i> | 51.33                                                          | 5.58 | 6.66 | 0.11 | 1.09     | 0.54     |                     |         | "bc"                   |
| <i>Funneliformis mosseae</i>    | 45.34                                                          | 5.05 | 0.00 | 0.11 | −1.75    | 3.73     |                     |         | "b"                    |
| <i>Gigaspora margarita</i>      | 42.67                                                          | 4.35 | 6.67 | 0.10 | −0.54    | −1.49    |                     |         | "b"                    |
|                                 | Mycorrhizal colonization of the root system (M%)               |      |      |      |          |          |                     |         |                        |
| <i>Glomus</i> sp.1              | 2.26                                                           | 0.76 | 0.90 | 0.34 | 0.94     | −0.25    | 4.54                | 0.005   | "bc"                   |
| <i>Glomus</i> sp.2              | 0.43                                                           | 0.11 | 0.10 | 0.24 | 1.12     | 0.88     |                     |         | "a"                    |
| <i>Septoglomus constrictum</i>  | 1.50                                                           | 0.83 | 0.30 | 0.56 | 2.06     | 4.36     |                     |         | "ac"                   |
| <i>Rhizophagus intraradices</i> | 2.45                                                           | 1.46 | 2.20 | 0.60 | 0.13     | −2.09    |                     |         | "c"                    |
| <i>Funneliformis mosseae</i>    | 1.40                                                           | 0.84 | 0.43 | 0.60 | 1.76     | 3.37     |                     |         | "ac"                   |
| <i>Gigaspora margarita</i>      | 0.77                                                           | 0.12 | 0.20 | 0.15 | −0.59    | −2.85    |                     |         | "ab"                   |
|                                 | Mycorrhizal colonization of root fragments (m%)                |      |      |      |          |          |                     |         |                        |
| <i>Glomus</i> sp.1              | 3.99                                                           | 1.57 | 1.79 | 0.39 | 0.89     | 0.39     | 2.65                | 0.048   | "a"                    |
| <i>Glomus</i> sp.2              | 1.34                                                           | 0.34 | 0.50 | 0.26 | 0.24     | −1.52    |                     |         | "a"                    |
| <i>Septoglomus constrictum</i>  | 3.12                                                           | 1.59 | 0.48 | 0.51 | 2.11     | 4.55     |                     |         | "a"                    |
| <i>Rhizophagus intraradices</i> | 4.61                                                           | 2.49 | 3.00 | 0.54 | 0.20     | −0.92    |                     |         | "a"                    |
| <i>Funneliformis mosseae</i>    | 3.29                                                           | 2.53 | 0.85 | 0.77 | 2.03     | 4.29     |                     |         | "a"                    |
| <i>Gigaspora margarita</i>      | 1.81                                                           | 0.10 | 0.13 | 0.06 | −0.53    | −1.84    |                     |         | "a"                    |
|                                 | Abundance of arbuscules in the root cortex (A%)                |      |      |      |          |          |                     |         |                        |
| <i>Glomus</i> sp.1              | 0.05                                                           | 0.05 | 0.08 | 0.94 | 0.36     | −2.91    | 2.52                | 0.057   | "a"                    |
| <i>Glomus</i> sp.2              | 0.00                                                           | 0.00 | 0.00 | —    | —        | —        |                     |         | "a"                    |
| <i>Septoglomus constrictum</i>  | 0.01                                                           | 0.01 | 0.02 | 1.37 | 0.61     | −3.33    |                     |         | "a"                    |
| <i>Rhizophagus intraradices</i> | 0.05                                                           | 0.07 | 0.12 | 1.21 | 0.55     | −3.18    |                     |         | "a"                    |
| <i>Funneliformis mosseae</i>    | 0.01                                                           | 0.01 | 0.02 | 1.37 | 0.61     | −3.33    |                     |         | "a"                    |
| <i>Gigaspora margarita</i>      | 0.01                                                           | 0.01 | 0.02 | 1.37 | 0.61     | −3.33    |                     |         | "a"                    |
|                                 | Abundance of arbuscules in the mycorrhized parts of roots (a%) |      |      |      |          |          |                     |         |                        |
| <i>Glomus</i> sp.1              | 2.24                                                           | 2.14 | 3.31 | 0.95 | 0.35     | −2.08    | 1.85                | 0.142   | "a"                    |
| <i>Glomus</i> sp.2              | 0.00                                                           | 0.00 | 0.00 | —    | —        | —        |                     |         | "a"                    |
| <i>Septoglomus constrictum</i>  | 0.59                                                           | 0.82 | 1.28 | 1.39 | 0.74     | −2.60    |                     |         | "a"                    |
| <i>Rhizophagus intraradices</i> | 1.90                                                           | 2.23 | 3.15 | 1.17 | 0.84     | −0.87    |                     |         | "a"                    |
| <i>Funneliformis mosseae</i>    | 0.52                                                           | 0.71 | 1.28 | 1.37 | 0.61     | −3.32    |                     |         | "a"                    |
| <i>Gigaspora margarita</i>      | 0.88                                                           | 1.23 | 1.92 | 1.39 | 0.73     | −2.61    |                     |         | "a"                    |

<sup>A</sup> Descriptive statistics (SD: standard deviation, IQR: interquartile range, CV: coefficient of variation).<sup>B</sup> One-way AVOVA results are given as *F*: *F*-statistics with numerator and denominator degrees of freedom between parentheses, and *P*-values.<sup>C</sup> The last column displays results of Tukey's honest significant difference (HSD) tests where the same letters of AMF species are not significantly different at  $P > 0.05$ .

It was the same for the weight gain of the fresh root material which averaged between 435 g in *F. mosseae* and 1248 g in *Glomus* sp.1. The average of RFW/SFW ratio showed levels ranging from 0.76 in *G. margarita* to 1.08 in *Glomus* sp.1, but which did not differ between AMF species nor from the control which averaged 0.94. Mean values for mycorrhizal responsiveness ranged between 0.77 in *F. mosseae* and 0.91 in *Glomus* sp.1. The average number of leaves per plantlet ranged from 95 in *Glomus* sp.2–221.4 in *Glomus* sp.1; the controls had only 67.6 leaves/plantlet. Finally, the number of nodes also improved significantly in inoculated plantlets; it varied between 156.2 nodes in *Glomus* sp.2 and 281 in *Glomus* sp.1. While it was only 111 nodes/plantlet in AMF-free plantlets (Fig. 1).

Pearson's correlation tests between plantlet growth traits revealed significant positive correlations ( $P < 0.001$ ) between almost all traits except for SRWC and RFW/SFW ratio (Fig. 2). In other terms, while the increases of all growth and morphometric traits go in the same way, SRWC and RFW/SFW ratio deemed constant, not-related and independent of the increases of the previous traits. Out of 36 correlations tests, two were negative and non-significant, the first was recoded between RFW/SFW and SFW, and the second between RFW/SFW and SDW.

### 3.3. Effects of mycorrhizal parameters on plantlet growth traits

The existing relationships between growth traits and the different mycorrhizal colonization parameters obtained in the olive plants

inoculated with different AMF strains were demonstrated using GLMs summarized in Table 2. These GLMs enabled the understanding of growth variation according to mycorrhization parameters determined at harvest after two-years growth. Overall, and for all the effects tested (with the exception of *A*% and *a*% on RFW/SFW ratio), plantlet growth was deemed positively correlated with the increase of mycorrhizal parameters (Fig. 3). Indeed, GLMs revealed that the five mycorrhizal parameters (*F*%, *M*%, *m*%, *A*%, *a*%) have positive and significant effects on the growth of the five morphometric and growth traits SH, NL, NN, SFW, and SDW. However, the models indicated no significant effects ( $P > 0.05$ ) for all mycorrhizal parameters on the variation of SRWC and RFW/SFW ratio. While the GLMs showed that the increase of *F*%, *M*%, *m*% induced a significant increase in the growth of RFW and values of MR (Table 2). Among the significant effects recorded for mycorrhizal parameters, the frequency of mycorrhizal colonization '*F*%' had the highest effects ( $P < 0.001$ ) on plantlet growth, especially for SFW ( $F = 30.39$ ,  $P < 0.001$ ) and NL ( $\chi^2 = 345$ ,  $P < 0.001$ ).

## 4. Discussion

The results obtained revealed that all the inoculated olive plantlets showed an improvement in the different growth parameters with the presence of colonization in their roots. This shows the endomycorrhizal effectiveness of the olive tree, and reveals the infectivity and effectiveness of all the AMF strains tested. Long et al. (2010), demonstrated that the strains of *G. margarita*, *R. intraradices* and *F. mosseae* are



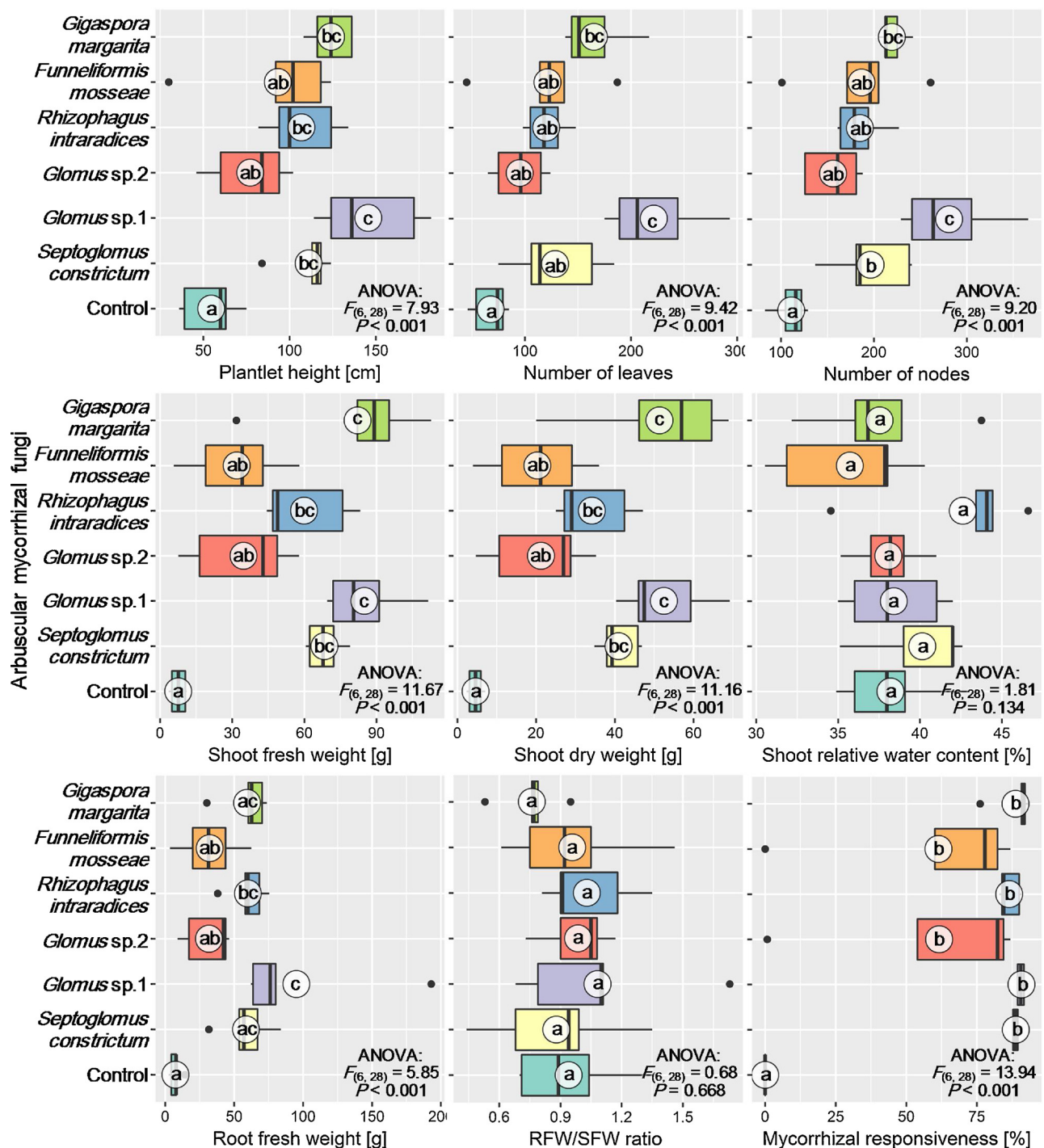


Fig. 1. Box plots representing morphometrics and growth traits of two-year-old olive plantlets inoculated using native and commercial AMF and control plantlets. Solid white circles indicate the means, whereas black circles are outliers. One-way AVOVA results are given as  $F$ :  $F$ -statistics with numerator and denominator degrees of freedom between parentheses, and  $P$ :  $P$ -values. Letters associated to white circles are results of Tukey's HSD tests, where the same letters are not significantly different at  $P > 0.05$ .

infective and effective in *Zinnia elegans* (Asteraceae). In previous studies, *S. constrictum* was demonstrated to be an effective mycorrhizal inoculum (Gong et al., 2013), and best adapted to germinate and colonize plant roots under experimental conditions (Marulanda et al., 2009). The results of this study are also in agreement with findings of Estaún et al. (2003); Meddad-Hamza (2010), which find that *R. intraradices*, *S. constrictum* and *F. mosseae* can effectively colonize olive roots. Regarding the positive effect of motorization on olive tree growth, our findings are consistent with the results obtained by

Caravaca et al. (2003a) and Calvente et al. (2004). Indeed, mycorrhizae play a vital role as growth agents in many fruit trees as a result of improved photosynthesis and water use (Bâ et al., 2000; Birhane et al., 2012).

Regardless of the AMF spore inoculated within the olive tree, the mycorrhizal plantlets grow better than the control plantlets. Mycorrhizal root colonization induces an increase in surface uptake, a larger area of explored soil, greater longevity of the absorbent roots, better utilization of less-available nutrients, and better retention of

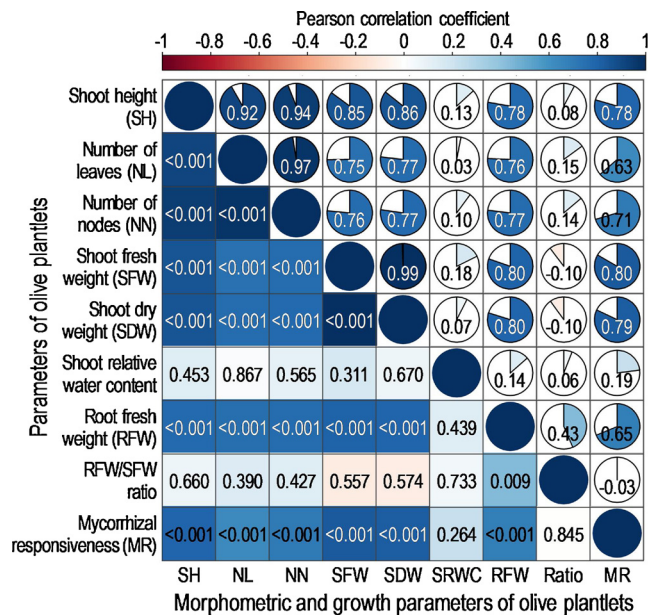


Fig. 2. Relationships between morphometric and growth parameters of olive plantlets inoculated using native and commercial AMF. Correlation test values are represented as Pearson correlation coefficients (shown by color and intensity of shading in pie charts and squares, and values above diagonal) and P-values (below diagonal). Strong correlations ( $r > 0.75$ ) are indicated in white font color.

soluble nutrients, thus reducing the reaction with soil colloids or leaching losses (Selvaraj and Chellappan, 2006; Neffar et al., 2015).

Our findings are consistent with those found by Meddad-Hamza (2010) that demonstrated better growth levels in olive trees inoculated with the fungal strains of *R. intraradices* and *F. mosseae*. However, the

results indicate a better responsiveness and better effect of the *R. intraradices* species compared to *F. mosseae*. In fact, different growth responses of the olive tree have been demonstrated following its inoculation with different AMF strains (Vitagliano and Citeresi, 1999; Calvente et al., 2004; Castillo et al., 2006; Meddad-Hamza, 2010); with *R. intraradices* being one of the most effective AMF in improving plant growth in terms of soil water absorption (Estaún et al., 2003; Marulanda et al., 2003).

The mycorrhizal responsiveness 'MR' (known also as mycorrhizal efficiency or also plant responsiveness, see Janos, 2007) determines the importance of the relationship between the host plant and fungi. This parameter indicates how plants benefit from AMF and can experience maximum growth. Knowing the MR of the host plant species is essential for predicting the host response to inoculation tests with AMF. MR of a given plant is not only determined by the host traits, but also by the fungal symbiont and environmental conditions (Estaún et al., 2010; Bradai et al., 2014, 2015).

Castillo et al. (2006) studied the effects of simple and multiple inoculation of the olive varieties Arbequina and Picual with AMF *R. intraradices*, *F. mosseae* and *Septoglomus viscosum* (syn. *Glomus viscosum*). They concluded that pre-inoculation of olive trees with AMF improves the health and vigor of plants and reduces the severity of the gall caused by nematodes in the roots. The remarkable effect of the fungal strain *Glomus* sp.1, compared to other tested AMF strains, on growth parameters and colonization parameters, may suggest a kind of adaptation and affinity of the Ferkani variety with respect to this fungal strain extracted from its natural rhizosphere. In fact, AMF isolates are often more infectious when used in the soil from which they were collected (Caravaca et al., 2003b). However, differences in levels of colonization efficiency of the different AMF strains used, can be attributed to (i) differences between fungi in the rate and extent of colonization, (ii) affinity and compatibility between the fungus and the host plant, or (iii) the conditions of the experiment.

In practice, it is very difficult to perform adequate controls in

Table 2

Generalized linear models testing the effects of mycorrhizal parameters on the variation of growth traits of olive plantlets (var. Ferkani) inoculated with native and commercial arbuscular mycorrhizal fungi. In GLM computations, the Poisson distribution and log link was fitted to NL and NN data, whereas Gaussian distributions and identity link were fitted to the other plant traits (SH, SFW, SDW, SRWC, RFW, RFW/SFW ratio, and MR).

| Mycorrhizal parameters * | Shoot height (SH) |         | Number of leaves (NL) |         | Number of nodes (NN) |         |
|--------------------------|-------------------|---------|-----------------------|---------|----------------------|---------|
|                          | $F_{(1,33)}$      | P       | $\chi^2$              | P       | $\chi^2$             | P       |
| F%                       | 27.92             | < 0.001 | 345.0                 | < 0.001 | 314.1                | < 0.001 |
| M%                       | 8.99              | 0.005   | 108.1                 | < 0.001 | 117.1                | < 0.001 |
| m%                       | 7.23              | 0.011   | 80.6                  | < 0.001 | 99.0                 | < 0.001 |
| A%                       | 4.97              | 0.033   | 60.6                  | < 0.001 | 50.2                 | < 0.001 |
| a%                       | 4.81              | 0.035   | 66.2                  | < 0.001 | 43.9                 | < 0.001 |

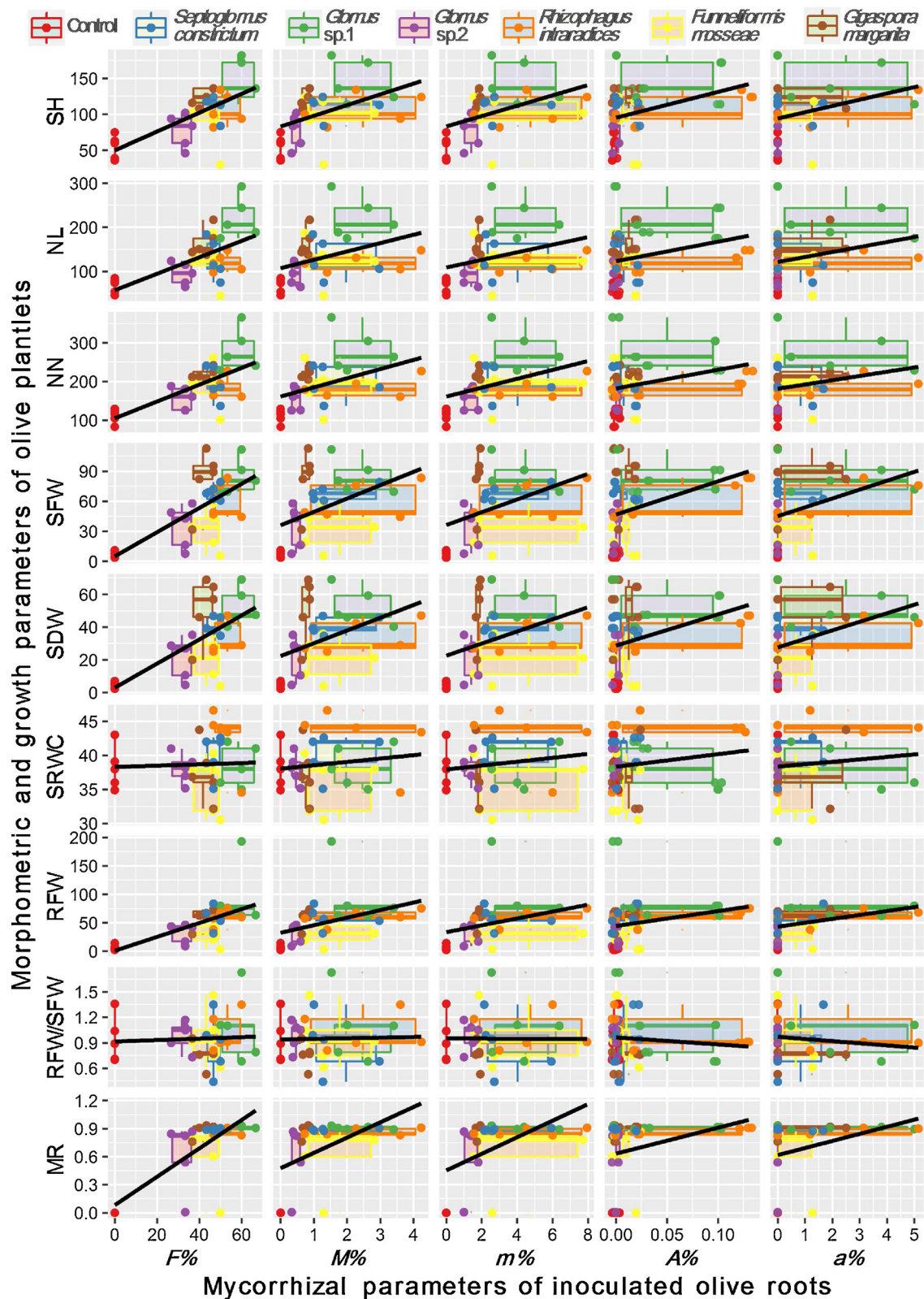
  

| Mycorrhizal parameters | Shoot fresh weight (SFW) |         | Shoot dry weight (SDW) |         | Shoot relative water content (SRWC) |       |
|------------------------|--------------------------|---------|------------------------|---------|-------------------------------------|-------|
|                        | $F_{(1,33)}$             | P       | $F_{(1,33)}$           | P       | $F_{(1,33)}$                        | P     |
| F%                     | 30.39                    | < 0.001 | 29.22                  | < 0.001 | 0.07                                | 0.798 |
| M%                     | 8.83                     | 0.005   | 7.73                   | 0.009   | 0.68                                | 0.416 |
| m%                     | 6.96                     | 0.013   | 6.00                   | 0.020   | 0.81                                | 0.374 |
| A%                     | 5.25                     | 0.029   | 4.51                   | 0.041   | 1.00                                | 0.325 |
| a%                     | 6.35                     | 0.017   | 5.72                   | 0.023   | 0.59                                | 0.446 |

| Mycorrhizal parameters | Root fresh weight (RFW) |         | RFW/SFW ratio |       | Mycorrhizal responsiveness (MR) |         |
|------------------------|-------------------------|---------|---------------|-------|---------------------------------|---------|
|                        | $F_{(1,33)}$            | P       | $F_{(1,33)}$  | P     | $F_{(1,33)}$                    | P       |
| F%                     | 22.06                   | < 0.001 | 0.11          | 0.746 | 53.27                           | < 0.001 |
| M%                     | 6.83                    | 0.013   | 0.03          | 0.875 | 11.53                           | 0.002   |
| m%                     | 4.80                    | 0.036   | 0.00          | 0.974 | 12.42                           | 0.001   |
| A%                     | 2.49                    | 0.124   | 0.38          | 0.543 | 2.80                            | 0.104   |
| a%                     | 2.87                    | 0.100   | 0.59          | 0.449 | 3.52                            | 0.069   |

\* (F%: frequency of mycorrhizal colonization, A%: abundance of arbuscules in the whole root cortex, a%: abundance of arbuscules in the mycorrhized parts of root fragments, M%: mycorrhizal colonization of the whole root cortex, m%: mycorrhizal colonization of root fragments).



**Fig. 3.** Relationship between mycorrhizal parameters and growth traits of olive plantlets inoculated using commercial and native AMF isolated from drylands of Algeria. The solid black lines represent a linear regression with a generalized linear model fit. Colored boxplots summarize the scattered dots for each AMF inoculum. (F%: frequency of mycorrhizal colonization, A%: abundance of arbuscules in the whole root cortex, a%: abundance of arbuscules in the mycorrhizal parts of root fragments, M%: mycorrhizal colonization of the whole root cortex, m%: mycorrhizal colonization of root fragments).



mycorrhizal studies, because the removal of mycorrhizal fungi causes changes in the chemical, biological, and physical properties of the soil, and inoculation with fungi is likely to introduce other organisms (Koide and Li, 1989). However, the impact of microbial factors or changes in soil fertility, as a result of sterilization, on plant growth is most often low compared to the impact of mycorrhizal treatments. Several studies show that plants and fungi differ in their composition and physiology (van der Heijden and Sanders, 2003v), which act directly on the symbiotic relationships between AMF and host plants. They also demonstrated the effect of AMF physiological diversity by comparing the responses of different plant species to fungal strains, and concluded that there exist variations between taxa and the intra-specific variability within AMF species in their capacity to promote plant growth (Stahl et al., 1998). These variations can also be manifested at the varietal level in the same plant species (Estaún et al., 2010; Meddad-Hamza et al., 2017). In addition, the geographical provenance of plant material plays a key-role in plantlet growth differences within the same species (Bezzalla et al., 2018).

The positive significant relationships found between plant productivity and AMF infection parameters may be a consequence of the functional diversity of AMF (van der Heijden and Sanders, 2003v). This can be explained by the difference in the mycorrhizal response that may have different patterns within the same plant species for the same fungal strain. Indeed, it is difficult to determine the similarities, differences and variations in the behavior of different plant species and even cultivars with respect to mycorrhizal symbiosis (Estaún et al., 2010). For instance, Gao et al. (2007) studied six varieties of rice inoculated with two different AMF, *Funneliformis mosseae* and *Claroideoglomus etunicatum* (syn. *Glomus uncatum*). They found that all but one variety showed better growth with *F. mosseae*; while *C. etunicatum* improved seedling growth of only four varieties. These data indicate that plant species may respond differently to dependency and non-dependency responsiveness factors.

In our experiment, we found that the local *Glomus* sp.1 and the commercial *Rhizophagus intraradices* were the most effective AMF, followed by, *F. mosseae*, *S. constrictum*, *G. margarita* and then *Glomus* sp.2. Our results are very consistent with those of Estaún et al. (2003); Veresoglou et al. (2011) and Li et al. (2014), which states that *R. intraradices* is more effective and acts better on growth improvement plants in particular annuals, than *F. mosseae* and *G. margarita*. This improvement of growth is the result of improved mineral nutrition (Veresoglou et al., 2011). The benefits of AMF to plant communities were quantified by measuring the impact of temporary suppression of these fungi by the application of fungicides (Abd-Alla et al., 2000). Obviously, when mycorrhizal formation is inhibited, significant reduced yields have been observed in some plant species (Lapointe and Molard, 1997).

## 5. Conclusion

Controlled inoculation of olive plantlets var. Ferkani with local AMF strains from the natural habitat of the variety and commercialized AMF species, showed that local strains, especially *Glomus* sp.1, compared to commercially available strains, are infective and rapidly colonize the root system of the inoculated plantlets. The effects of inoculations on the increase of different parameters of plantlet growth are highly significant. The results of this controlled mycorrhization indicate that these fungal partners do not seem to be specific, even if there is a variability between the colonization rates recorded in the olive tree. Moreover, the olive variety Ferkani can establish symbiosis with several AMF species that differ phylogenetically and ecologically, and therefore, Ferkani should be less-specific to AMF in field conditions. This study demonstrates the mycorrhizal effectiveness of local AMF species and suggests promising prospects for their application.

## Funding

This study was not funded by any sources.

## Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Authors' contributions

HC and MNM contributed equally to this work. AB conceived the ideas and designed methodology. All authors have read and approved the manuscript.

## Declaration of Competing Interest

The authors declare no conflict of interest.

## Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.scienta.2019.108969>.

## References

- Abd-Alla, M.H., Omar, S.A., Karanxha, S., 2000. The impact of pesticides on arbuscular mycorrhizal and nitrogen-fixing symbioses in legumes. *Appl. Soil Ecol.* 14 (3), 191–200. [https://doi.org/10.1016/S0929-1393\(00\)00056-1](https://doi.org/10.1016/S0929-1393(00)00056-1).
- Akhtar, M.S., Siddiqui, Z.A., 2008. Arbuscular mycorrhizal fungi as potential bioprotectants against plant pathogens. In: Siddiqui, Z.A., Akhtar, M.S., Futai, K. (Eds.), *Mycorrhizae: Sustainable Agriculture and Forestry*. Springer, Netherlands, pp. 61–97. [https://doi.org/10.1007/978-1-4020-8770-7\\_3](https://doi.org/10.1007/978-1-4020-8770-7_3).
- Aktar, W., Sengupta, D., Chowdhury, A., 2009. Impact of pesticides use in agriculture: their benefits and hazards. *Toxicology* 2 (1), 1–12. <https://doi.org/10.2478/v10102-009-0001-7>.
- Bâ, M., Plenchette, C., Danthu, P., Duponnois, R., Guissou, T., 2000. Functional compatibility of two arbuscular mycorrhizae with thirteen fruit trees in Senegal. *Agrofor. Syst.* 50, 95–105. <https://doi.org/10.1023/a:1006482904452>.
- Benabderrahmane, M.C., Chenchouni, H., 2010. Assessing environmental sensitivity areas to desertification in eastern Algeria using Mediterranean desertification and land use “MEDALUS” model. *Int. J. Sustain. Water Environ. Syst.* 1, 5–10. <https://doi.org/10.5383/swes.01.01.002>.
- Bezzalla, A., Boudjabi, S., Chenchouni, H., 2018. Seedlings of Argan (*Argania spinosa*) from different geographical provenances reveal variable morphological growth responses to progressive drought stress under nursery conditions. *Agrofor. Syst.* 92, 1201–1211. <https://doi.org/10.1007/s10457-016-0057-2>.
- Birhane, E., Sterck, F.J., Fetene, M., Bongers, F., Kuyper, T.W., 2012. Arbuscular mycorrhizal fungi enhance photosynthesis, water use efficiency, and growth of frankincense seedlings under pulsed water availability conditions. *Oecologia* 169 (4), 895–904. <https://doi.org/10.1007/s00442-012-2258-3>.
- Blondel, J., Aronson, J., Bodiou, J.Y., Boeuf, G., 2010. *The Mediterranean Region: Biological Diversity in Space and Time*. Oxford University Press.
- Bouaroudj, S., Menad, A., Bounamous, A., Ali-Khodja, H., Gherib, A., Weigel, D.E., Chenchouni, H., 2019. Assessment of water quality at the largest dam in Algeria (Bni Haroun Dam) and effects of irrigation on soil characteristics of agricultural lands. *Chemosphere* 219, 76–88. <https://doi.org/10.1016/j.chemosphere.2018.11.193>.
- Boudjabi, S., Kribaa, M., Chenchouni, H., 2015. Growth, physiology and yield of durum wheat (*Triticum durum*) treated with sewage sludge under water stress conditions. *EXCLI J.* 14, 320–334. <https://doi.org/10.17179/excli2014-715>.
- Boudjabi, S., Kribaa, M., Chenchouni, H., 2019. Sewage sludge fertilization alleviates drought stress and improves physiological adaptation and yield performances in Durum Wheat (*Triticum durum*): a double-edged sword. *J. King Saud Univ. Sci.* 31, 336–344. <https://doi.org/10.1016/j.jksus.2017.12.012>.
- Boughalleb, F., Hajlaoui, H., 2011. Physiological and anatomical changes induced by drought in two olive cultivars (cv Zalmati and Chemlali). *Acta Physiol. Plant.* 33, 53–65. <https://doi.org/10.1007/s11738-010-0516-8>.
- Bradai, L., Bissati, S., Chenchouni, H., 2014. Desert truffles of the North Algerian Sahara: diversity and bioecology. *Emir. J. Food Agric.* 26 (5), 425–435. <https://doi.org/10.9755/efja.v26i5.16520>.
- Bradai, L., Bissati, S., Chenchouni, H., Amrani, K., 2015. Effects of climate on the productivity of desert truffles beneath hyper-arid conditions. *Int. J. Biometeorol.* 59 (7), 907–915. <https://doi.org/10.1007/s00484-014-0891-8>.
- Brundrett, M.C., 1991. Mycorrhizas in natural ecosystems. *Adv. Ecol. Res.* 21, 171–213. [https://doi.org/10.1016/S0065-2504\(08\)60099-9](https://doi.org/10.1016/S0065-2504(08)60099-9).
- Brundrett, M.C., Piche, Y., Peterson, R.L., 1984. A new method for observing the morphology of vesicular-arbuscular mycorrhizae. *Can. J. Bot.* 62, 2128–2134. <https://doi.org/10.1139/b84-290>.



- Calvente, R., Cano, C., Ferrol, N., Azcón-Aguilar, C., Barea, J.M., 2004. Analysing natural diversity of arbuscular mycorrhizal fungi in olive tree (*Olea europaea* L.) plantations and assessment of the effectiveness of native fungal isolates as inoculants for commercial cultivars of olive plantlets. *Appl. Soil Ecol.* 26, 11–19. <https://doi.org/10.1016/j.apsoil.2003.10.009>.
- Caravaca, F., Díaz, E., Barea, J.M., Azcón-Aguilar, C., Roldán, A., 2003a. Photosynthetic and transpiration rates of *Olea europaea* subsp. *sylvestris* and *Rhamnus lycioides* as affected by water deficit and mycorrhiza. *Biol. Plant.* 46, 637–639. <https://doi.org/10.1023/A:1024880121096>.
- Caravaca, F., Barea, J.M., Palenzuela, J., Figueroa, D., Alguacil, M.M., Roldán, A., 2003b. Establishment of shrub species in a degraded semiarid site after inoculation with native or allochthonous arbuscular mycorrhizal fungi. *Appl. Soil Ecol.* 22 (2), 103–111. [https://doi.org/10.1016/S0929-1393\(02\)00136-1](https://doi.org/10.1016/S0929-1393(02)00136-1).
- Carvalho, F.P., 2017. Pesticides, environment, and food safety. *Food Energy Secur.* 6 (2), 48–60. <https://doi.org/10.1002/fes3.108>.
- Castillo, P., Nico, A.I., Azcón-Aguilar, C., Del Río Rincón, C., Calvet, C., Jiménez-Díaz, R.M., 2006. Protection of olive planting stocks against parasitism of root-knot nematodes by arbuscular mycorrhizal fungi. *Plant Pathol.* 55 (5), 705–713. <https://doi.org/10.1111/j.1365-3059.2006.01400.x>.
- Chafai, S., Mimeche, F., Chenchouni, H., 2019. Diversity of insects associated with olive (*Oleaceae*) groves across a dryland climate gradient in Algeria. *Can. Entomol.* 151, 629–647. <https://doi.org/10.4039/tce.2019.35>.
- Chenchouni, H., 2010. Drought-induced mass mortality of Atlas Cedar forest (*Cedrus atlantica*) in Algeria. In: Parrota, J.A., Carr, M.A. (Eds.), *The International Forestry Review*, 33th IUFRO World Congress. 23–28 August 2010, Seoul, Korea.
- Djellab, S., Mebarkia, N., Neffar, S., Chenchouni, H., 2019. Diversity and phenology of hoverflies (Diptera: Syrphidae) in pine forests (*Pinus halepensis* Miller) of Algeria. *J. Asia Pac. Entomol.* 22, 766–777. <https://doi.org/10.1016/j.aspen.2019.05.012>.
- Estaún, V., Calvet, C., Cambrubí, A., 2010. Effect of differences among crop species and cultivars on the arbuscular mycorrhizal symbiosis. In: Koltai, H., Kapulnik, Y. (Eds.), *Arbuscular Mycorrhizas: Physiology and Function*. Springer, Berlin, pp. 279–295. [https://doi.org/10.1007/978-90-481-9489-6\\_13](https://doi.org/10.1007/978-90-481-9489-6_13).
- Estaún, V., Cambrubí, A., Calvet, C., Pinochet, J., 2003. Nursery and field response of olive trees inoculated with two arbuscular mycorrhizal fungi, *Glomus intraradices* and *Glomus mosseae*. *J. Am. Soc. Hortic. Sci.* 128, 767–775. <https://doi.org/10.21273/jashs.128.5.0767>.
- Gao, X., Kuyper, T.W., Zou, C., Zhang, F., Hoffland, E., 2007. Mycorrhizal responsiveness of aerobic rice genotypes is negatively correlated with their zinc uptake when non-mycorrhizal. *Plant Soil* 290 (1–2), 283–291. <https://doi.org/10.1007/s11104-006-9160-x>.
- Garg, N., Chandel, S., 2010. Arbuscular mycorrhizal networks: process and functions. A review. *Agron. Sustain. Dev.* 30, 581–599. <https://doi.org/10.1051/agro/2009054>.
- Ghazi, N., Al-Karak, G.N., 2006. Nursery inoculation of tomato with arbuscular mycorrhizal fungi and subsequent performance under irrigation with saline water. *Sci. Hortic.* 109, 1–7. <https://doi.org/10.1016/j.scienta.2006.02.019>.
- Giri, B., Kapoor, R., Mukerji, K.G., 2007. Improved tolerance of *Acacia nilotica* to salt stress by arbuscular mycorrhiza, *Glomus fasciculatum* may be partly related to elevated K/Na ratios in root and shoot tissues. *Microbial. Ecol.* 54, 753–760. <https://doi.org/10.1007/s00248-007-9239-9>.
- Gong, M., Tang, M., Chen, H., Zhang, Q., Feng, X., 2013. Effects of two *Glomus* species on the growth and physiological performance of *Sophora davidii* seedlings under water stress. *New For.* 44 (3), 399–408. <https://doi.org/10.1007/s11056-012-9349-1>.
- Harrier, L.A., Watson, C.A., 2004. The potential role of arbuscular mycorrhizal (AM) fungi in the bioprotection of plants against soil-borne pathogens in organic and/or other sustainable farming systems. *Pest Manage. Sci.* 60, 149–157. <https://doi.org/10.1002/ps.820>.
- Helgason, T., Fitter, A., 2009. Natural selection and the evolutionary ecology of the arbuscular mycorrhizal fungi (Phylum Glomeromycota). *J. Exp. Bot.* 60, 2465–2480. <https://doi.org/10.1093/jxb/erp144>.
- Hernández, A.F., Parrón, T., Tsatsakis, A.M., Requena, M., Alarcón, R., López-Guarnido, O., 2013. Toxic effects of pesticide mixtures at a molecular level: their relevance to human health. *Toxicology* 307, 136–145. <https://doi.org/10.1016/j.tox.2012.06.009>.
- Hillocks, R.J., 2012. Farming with fewer pesticides: EU pesticide review and resulting challenges for UK agriculture. *Crop. Prot.* 31 (1), 85–93. <https://doi.org/10.1016/j.cropro.2011.08.008>.
- INRAA (Institut National de la Recherche Agronomique d'Algérie), 2006. Rapport national sur l'état des ressources phytogénétiques pour l'alimentation et l'agriculture – Algérie. Deuxième rapport national sur l'état des ressources phytogénétiques. Institut National de la Recherche Agronomique d'Algérie (INRAA), El Harrach, Algeria. <http://www.fao.org/docrep/013/11500e/Algeria.pdf>.
- ITAFV (Institut technique d'arboriculture fruitière et de la vigne), 2013. 1er Salon international de l'oléiculture (2013). Algiers, Algeria.
- Janos, D.P., 2007. Plant responsiveness to mycorrhizas differs from dependence upon mycorrhizas. *Mycorrhiza* 17, 75–91. <https://doi.org/10.1007/s00572-006-0094-1>.
- Johansson, J.F., Paul, L.R., Finlay, R.D., 2004. Microbial interactions in the mycorrhizosphere and their significance for sustainable agriculture. *FEMS Microbiol. Ecol.* 48 (1), 1–13. <https://doi.org/10.1016/j.femsec.2003.11.012>.
- Kleijn, D., Berendse, F., Smit, R., Gilissen, M., Smit, J., Brak, B., Groeneweld, R., 2004. Ecological effectiveness of agri-environment schemes in different agricultural landscapes in the Netherlands. *Conserv. Biol.* 18 (3), 775–786. <https://doi.org/10.1111/j.1523-1739.2004.00550.x>.
- Koide, R.T., Li, M., 1989. Appropriate controls for vesicular arbuscular mycorrhiza research. *New Phytol.* 111, 35–44. <https://doi.org/10.1111/j.1469-8137.1989.tb04215.x>.
- Lapointe, L., Molard, J., 1997. Costs and benefits of mycorrhizal infection in a spring ephemeral, *Erythronium americanum*. *New Phytol.* 135, 491–500. <https://doi.org/10.1046/j.1469-8137.1997.00672.x>.
- Li, T., Lin, G., Zhang, X., Chen, Y., Zhang, S., Chen, B., 2014. Relative importance of an arbuscular mycorrhizal fungus (*Rhizophagus intraradices*) and root hairs in plant drought tolerance. *Mycorrhiza* 24 (8), 595–602. <https://doi.org/10.1007/s00572-014-0578-3>.
- Long, L.K., Yao, Q., Huang, Y.H., Yang, R.H., Guo, J., Zhu, H.H., 2010. Effects of arbuscular mycorrhizal fungi on zinnia and the different colonization between *Gigaspora* and *Glomus*. *World J. Microbiol. Biotechnol.* 26 (8), 1527–1531. <https://doi.org/10.1007/s11274-010-0313-y>.
- Marulanda, A., Azcón, R., Ruiz-Lozano, J.M., 2003. Contribution of six arbuscular mycorrhizal fungal isolates to water uptake by *Lactuca sativa* plants under drought stress. *Physiol. Plant.* 119, 526–533. <https://doi.org/10.1046/j.1399-3054.2003.00196.x>.
- Marulanda, A., Barea, J.M., Azcón, R., 2009. Stimulation of plant growth and drought tolerance by native microorganisms (AM fungi and bacteria) from dry environments: mechanisms related to bacterial effectiveness. *J. Plant Growth Regul.* 28 (2), 115–124. <https://doi.org/10.1007/s00344-009-9079-6>.
- Meddad-Hamza, A., 2010. Les mycorhizes de l'olivier *Olea europaea* L. aspects écologiques, effet sur la croissance et exploitation en pépinière. Doctoral thesis, Univ. Annaba, Algeria.
- Meddad-Hamza, A., Hamza, N., Neffar, S., Beddier, A., Gianinazzi, S., Chenchouni, H., 2017. Spatiotemporal variation of arbuscular mycorrhizal fungal colonization in olive (*Olea europaea* L.) roots across a broad mesic-xeric climatic gradient in North Africa. *Sci. Total Environ.* 583C, 176–189. <https://doi.org/10.1016/j.scitotenv.2017.01.049>.
- Mekahlia, M.N., Beddier, A., Chenchouni, H., 2013. Mycorrhizal dependency in the olive tree (*Olea europaea*) across a xeric climatic gradient. *Adv. Environ. Biol.* 7, 2166–2174.
- Mekahlia, M.N., Tine, S., Menasria, T., Amieur, H., Salhi, H., 2016. In vitro biomarker responses of earthworm *Lumbricus terrestris* exposed to herbicide sekator and phosphate fertilizer. *Water Air Soil Pollut.* 227 (1), 15. <https://doi.org/10.1007/s11270-015-2712-z>.
- Miransari, M., Bahrami, H.A., Rejali, F., Malakouti, M.J., 2008. Using arbuscular mycorrhiza to alleviate the stress of soil compaction on wheat (*Triticum aestivum* L.) growth. *Soil Biol. Biochem.* 40 (5), 1197–1206. <https://doi.org/10.1016/j.soilbio.2007.12.014>.
- Neffar, S., Beddier, A., Chenchouni, H., 2015. Effects of soil chemical properties and seasonality on mycorrhizal status of prickly pear (*Opuntia ficus-indica*) planted in hot arid steppe rangelands. *Sains Malays.* 44, 671–680. <https://doi.org/10.17576/jsm-2015-4405-05>.
- Neffar, S., Menasria, T., Chenchouni, H., 2018. Diversity and functional traits of spontaneous plant species in Algerian rangelands rehabilitated with prickly pear (*Opuntia ficus-indica* L.) plantations. *Turk. J. Bot.* 42 (4), 448–461. <https://doi.org/10.3906/bot-1801-39>.
- Nelsen, C.E., 1987. The water relations of vesicular-arbuscular mycorrhizal systems. In: Safir, G.R. (Ed.), *Ecophysiology of VA Mycorrhizal Plants*. CRC Press, Boca Raton, Florida, pp. 71–79.
- Oehl, F., Jansa, J., Ineichen, K., Mäder, P., van der Heijden, M.G.A., 2011b. Arbuscular Mycorrhizal Fungi As Bio-indicators in Swiss Agricultural Soils. *FAO* 2011b.
- Oehl, F., Silva, G.A., Goto, B.T., Sieverding, E., 2011a. Glomeromycota: three new genera and glomoid species reorganized. *Mycotaxon* 116, 75–120. <https://doi.org/10.5248/116.75>.
- Oustani, M., Halilat, M.T., Chenchouni, H., 2015. Effect of poultry manure on the yield and nutrients uptake of potato under saline conditions of arid regions. *Emir. J. Food Agric.* 27 (1), 106–120. <https://doi.org/10.9755/ejfa.v27i1.17971>.
- Parniske, M., 2008. Arbuscular mycorrhiza: the mother of plant root endosymbioses. *Nat. Rev. Microbiol.* 6, 763–775. <https://doi.org/10.1038/nrmicro1987>.
- Porrás-Soriano, A., Soriano-Martín, M.L., Porrás-Piedra, A., Azcón, R., 2009. Arbuscular mycorrhizal fungi increased growth, nutrient uptake and tolerance to salinity in olive trees under nursery conditions. *J. Plant Physiol.* 166 (13), 1350–1359. <https://doi.org/10.1016/j.jplph.2009.02.010>.
- R Core Team, 2019. R: A Language and Environment for Statistical Computing. Available at: R Foundation for Statistical Computing, Vienna, Austria. <http://www.r-project.org>.
- Robert, V., Vu, D., Ben Hadj Amor, A., et al., 2013. MycoBank gearing up for new horizons. *IMA Fungus* 4, 371–379. <http://www.mycobank.org/>. <http://www.imafungus.org/Issue/42/26.pdf>.
- Ruiz-Lozano, J.M., Porcel, R., Bárzana, G., Azcón, R., Aroca, R., 2012. Contribution of arbuscular mycorrhizal symbiosis to plant drought tolerance: state of the art. In: Aroca, R. (Ed.), *Plant Responses to Drought Stress: from Morphological to Molecular Features*. Springer, Berlin Heidelberg, pp. 335–362. [https://doi.org/10.1007/978-3-642-32653-0\\_13](https://doi.org/10.1007/978-3-642-32653-0_13).
- Schreinemachers, P., Tipraqsa, P., 2012. Agricultural pesticides and land use intensification in high, middle and low income countries. *Food Policy* 37 (6), 616–626. <https://doi.org/10.1016/j.foodpol.2012.06.003>.
- Selvaraj, T., Chelleppan, P., 2006. Arbuscular mycorrhizae: a diverse personality. *Central Eur. J. Agric.* 7, 349–358. <https://hrack.srce.hr/17369>.
- Sivasithamparan, K., Dixon, K.W., Barrett, R.L., 2002. Microorganisms in Plant Conservation and Biodiversity. Springer Netherlands, Dordrecht, Netherlands, <https://doi.org/10.1007/0-306-48099-9>.
- Stahl, P.D., Frost, S.M., Williams, S.E., Schuman, G.E., 1998. Arbuscular mycorrhizae and water stress tolerance of Wyoming Big Sagebrush seedlings. *Soil Sci. Soc. Am. J.* 62, 1309–1313. <https://doi.org/10.2136/sssaj1998.03615995006200050023x>.
- Tattini, M., 1990. Effect of increasing nutrient concentration on growth and nitrogen uptake of container-grown peach and olive. In: In: van Beusichem, M.L. (Ed.), *Plant Nutrition — Physiology and Applications*. Developments in Plant and Soil Sciences,

- vol. 41. Springer, Dordrecht, pp. 515–518. [https://doi.org/10.1007/978-94-009-0585-6\\_85](https://doi.org/10.1007/978-94-009-0585-6_85).
- Trouvelot, A., Kough, J.L., Gianinazzi, V., 1986. Mesure de taux de mycorhization VA d'un système racinaire. Recherche de méthodes d'estimation ayant une signification fonctionnelle. In: Gianinazzi-Pearson, V., Gianinazzi, S. (Eds.), *Physiological and genetical Aspects of Mycorrhizae*. INRA, Paris, France, pp. 217–221.
- van der Heijden, M.G.A., Sanders, I.R., 2003v. *Mycorrhizal Ecology*. Springer <https://doi.org/10.1007/978-3-540-38364-2>.
- Veresoglou, S.D., Shaw, L.J., Sen, R., 2011. *Glomus intraradices* and *Gigaspora margarita* arbuscular mycorrhizal associations differentially affect nitrogen and potassium nutrition of *Plantago lanceolata* in a low fertility dune soil. *Plant Soil* 340, 481–490. <https://doi.org/10.1007/s11104-010-0619-4>.
- Vierheilig, H., Schweiger, P., Brundrett, M., 2005. An overview of methods for the detection and observation of arbuscular mycorrhizal fungi in roots. *Physiol. Plant.* 125, 393–404. <https://doi.org/10.1111/j.1399-3054.2005.00564.x>.
- Vitagliano, C., Citeresi, A.S., 1999. Plant growth of *Olea europaea* L. as influenced by arbuscular mycorrhizal fungi. *Acta Hort.* 474, 357–362. <https://doi.org/10.17660/ActaHortic.1999.474.74>.
- Wang, B., Qiu, Y.L., 2006. Phylogenetic distribution and evolution of mycorrhizas in land plants. *Mycorrhiza* 16, 299–363. <https://doi.org/10.1007/s00572-005-0033-6>.
- Wei, T., Simko, V., 2016. Corplot: Visualization of a Correlation Matrix. R package version 0.77,. <https://cran.r-project.org/package=corplot>.
- Xavier, I.J., Boyetchko, S.M., 2002. Arbuscular mycorrhizal fungi as biostimulants and bioprotectants of crops. In: Khachatourians, G.G., Arora, D.K. (Eds.), *Appl. Mycol. Biotechnol.*, Vol. 2: Agriculture and Food Production. Elsevier, Amsterdam, pp. 311–330. [https://doi.org/10.1016/S1874-5334\(02\)80015-6](https://doi.org/10.1016/S1874-5334(02)80015-6).